

Name: _____

Date: _____

Class: _____

MyDesign Portfolio

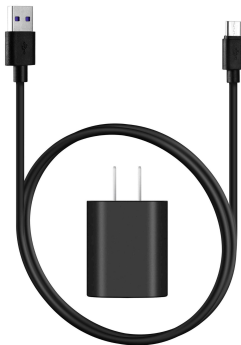
Element B Initial Template

Element B: Documentation and Analysis of Prior Design Attempts

Prior Design Attempt #1 - Anker PowerLine LED Charging Cable

Listing on Amazon

Image



Details

LED charging cables such as those sold by Anker contain embedded light strips that illuminate when the cable is actively transferring power. According to product descriptions on Amazon, the lights are powered directly by the electrical current flowing through the cable and are primarily marketed for aesthetic appeal and visibility during gaming or desk setups. The illumination occurs only while the cable is plugged into a powered source and actively charging a device. The materials used are standard PVC or braided nylon with integrated LED wiring.

Analysis

Name: _____

Date: _____

Class: _____

While the LED feature improves visibility when the charger is already connected to power, it does not help users locate the cable when it is unplugged. It also does not assist in locating the wall outlet itself. Because the LEDs require active electricity, this design does not solve the issue of finding a charger in a completely dark room before plugging it in. Therefore, it only partially addresses nighttime usability and fails to meet the full problem definition.

Prior Design Attempt #2 - Glow in the Dark Cable Covers

Sold through online retailers such as Etsy and Aliexpress

Image



Details

Glow in the dark cables and sleeves use phosphorescent materials that absorb ambient light and slowly release it over time. Product descriptions on Etsy and AliExpress state that the material must be exposed to light in order to “charge” before glowing in darkness. These accessories are passive systems, meaning they do not require electrical power to glow. However, brightness levels vary based on material quality and duration of prior light exposure. Many are thin outer coatings or removable sleeves placed over standard cables.

Analysis

This design improves nighttime visibility of the cable itself, but it does not help users locate the wall outlet. In addition, online product reviews frequently mention that the glow fades after a short period and may not remain bright for several hours. Durability is also limited because



Name: _____

Date: _____

Class: _____

external sleeves can crack or peel with repeated bending. As a result, this solution only addresses half of the defined problem and lacks long-term reliability.

Prior Design Attempt #3 - GE Plug In LED Night Light

Source

Manufacturer specifications and product descriptions from GE Lighting and retail listings from Walmart.

Image



Details

Plug-in LED night lights such as those manufactured by GE Lighting are designed to illuminate wall outlets and surrounding areas in low light conditions. Models include light sensors that automatically activate in darkness and deactivate in daylight. These devices plug directly into an outlet and provide continuous illumination to improve visibility.

Analysis

Although this solution makes the outlet easier to see, it requires the user to purchase and install a separate device. It also occupies outlet space that could otherwise be used for charging. Most importantly, it does not improve the visibility of the charger cable itself and does not provide directional guidance toward the outlet. Instead of redesigning the charger, this solution modifies the environment, meaning it does not directly solve the usability problem identified in Element A.

Rubric

Name: _____

Date: _____

Class: _____

Element B: Documentation and Analysis of Prior Design Attempts						
	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
B1. Documentation of plausible prior attempts to solve the problem and/or related problems is drawn from _____.	a wide array of clearly-identified and consistently-credible sources	a variety of clearly-identified and consistently-credible sources	several—but not necessarily varied—clearly-identified and generally-credible sources	a limited number of sources, some of which may not be clearly identified and/or credible	only one or two sources that may not be clearly identified and/or credible	missing or minimal (a single source that is not clearly identified and/or credible)
	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
B2. The analysis of past and current attempts to solve the problem — including both strengths and shortcomings — _____.	is consistently clear, detailed, and supported by relevant data	is clear, generally detailed, and supported by relevant data	is generally clear and contains some detail and relevant supporting data	is overly general and contains little detail and/or relevant supporting data	is vague and is missing any relevant details and/or relevant supporting data	cannot be inferred from information intended as analysis of past and/or current attempts to solve the problem